Lecture - 6

❖ Desk Checking:

Desk checking is a manual, non-computerized technique where a programmer traces through the logic of an algorithm or source code line by line using pen and paper.

- **Why Use Desk Checking?**
  - ✓ ***Catch Logic Errors Early:*** It excels at finding logic flaws (like off-by-one errors in loops or incorrect math formulas) that code compilers might miss.
  - ✓ ***Cost-Effective:*** Because it is performed during the design or early coding phase (often using pseudocode), it saves significant time and debugging resources later.
  - ✓ ***Deepens Understanding***: Forcing yourself to walk through a complex algorithm manually ensures you actually understand how the data flows.

- **Example:** Algorithm with a Hidden Bug

  Let's look at a loop that is supposed to calculate the sum of numbers from 1 up to a given number (N), where N = 3.The correct answer should be 6 (1 + 2 + 3).

```
Line 1: N = 3
Line 2: total = 0
Line 3: i = 1
Line 4: WHILE i < N
Line 5:     total = total + i
Line 6:     i = i + 1
Line 7: ENDWHILE
Line 8: OUTPUT total
```

The Desk Check Trace Table

By tracking the execution step-by-step, watch what happens to the output:

LineNumber	N	total	i	Condition(i<N)	Output
1 (init)	3	-	-	-	-
2 (init)	3	0	-	-	-
3 (init)	3	0	1	-	-
4 (Check)	3	0	1	1<3 (true)	-
5 (Add)	3	1	1	-	-
6 (Increment)	3	1	2	-	-
4 (Check)	3	1	2	2<3 (true)	-
5 (Add)	3	3	2	-	-
6 (Increment)	3	3	3	-	-
4 (Check)	3	3	3	3<3 (true)	-
8 (Output)	3	3	3	-	3

- The desk check reveals that the final output is 3, not the expected 6.
- Because Line 4 uses a strictly-less-than operator (i < N), the loop terminates early and completely skips adding the final number (3). To fix this bug, Line 4 needs to be changed to WHILE i <= N.